Java for beginners

Operators in Java



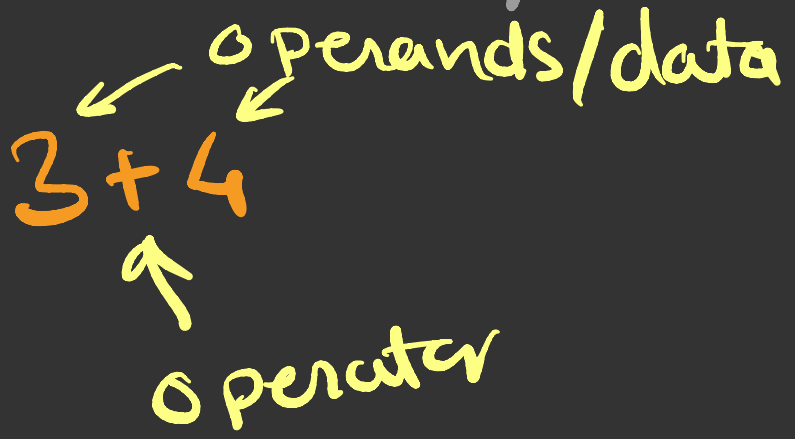
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Agenda

1. Classification of operators
2. Unary operators
3. Arithmetic operators
4. Bitwise Operators
5. Relational operators
6. Logical operators
7. Conditional Operator
8. Assignment operators
9. instance of operator

Classification of Operators

Operators are used to manipulate data to produce useful information



1. Unary operators
2. Arithmetic operators
3. Bitwise Operators
4. Relational Operators
5. Logical operators
6. Conditional Operator
7. Assignment operators
8. instance of operator

Unary Operators

+ , - , ++ , --

unary + unary -

+5 -7

Increment

++

```
int x = 5;
x++;  ⟶  x = x+1
++x;  ⟶  x = x+1
```

post increment → x++;

++x; → preincrement

Decrement

--

```
int x = 5;
x--;  ⟶  x = x-1
--x;  ⟶  x = x-1
```

post decrement: x--;

predecrement: --x;

increment & decrement operators will work only on variables and not constants.

Arithmetic Operators

*, /, %
+, −

a * b / c ✓

a / b * c ✓

a + b * c ✓

```
7 ) 6 ( 0
    0
    ---
    6
```

6 % 7 → 6

```
2 ) 5.5 ( 2
    4.0
    ---
    1.5
```

5 % 2 → 1

−5 % 2 → −1

5 % −2 → 1

−5 % −2 → −1

Unlike C, java allows real constants with modulo operator

Bitwise Operators

& , | , ^ , ~ , >> , <<

AND OR XOR NOT Right Shift Left Shift

Bit → 0 or 1

35 & 53 → 33

0 & 0 → 0
0 & 1 → 0
1 & 0 → 0
1 & 1 → 1

35 = 0010 0011
53 = 0011 0101

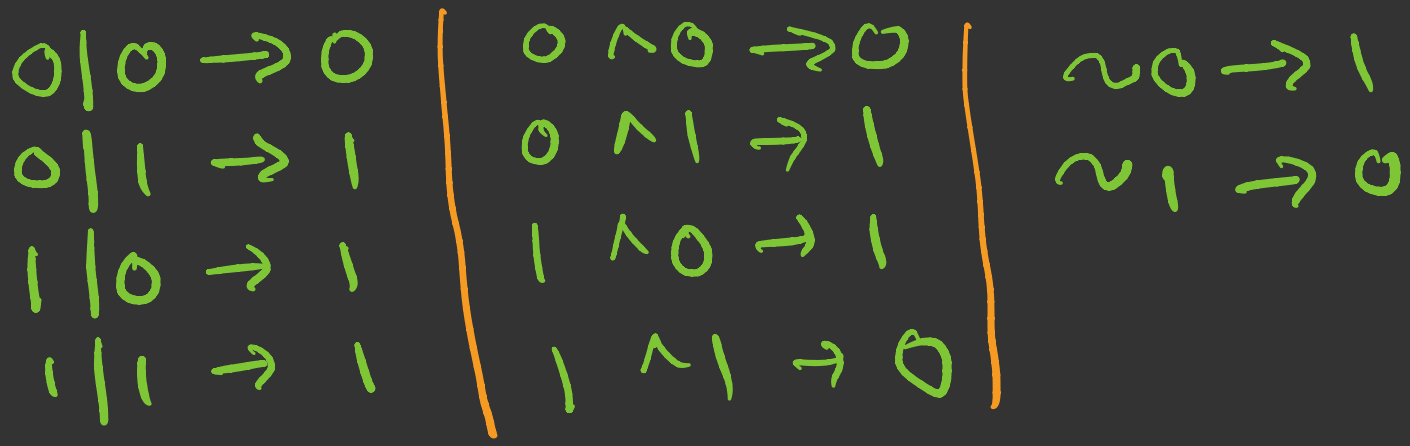
0010 0001

32 1

2^7 2^6 2^5 2^4 2^3 2^2 2^1 2^0

128 64 32 16 8 4 2 1

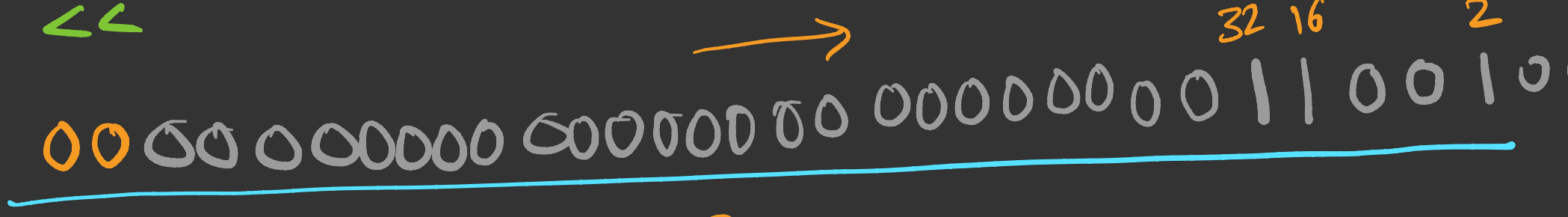
53 =


0 | 0 → 0
0 | 1 → 1
1 | 0 → 1
1 | 1 → 1
0 ^ 0 → 0
0 ^ 1 → 1
1 ^ 0 → 1
1 ^ 1 → 0
~0 → 1
~1 → 0

>>

<<

200 >> 2 → 50


32 16 2
00000000000000000000000011001000

12 << 3


00000000000000000000110000 = 96

Relational Operators

<, >, <=, >=

==, !=

- Always give boolean type result

int cannot be converted to boolean

Logical Operators

!	NOT
&&	AND
||	OR

! boolean

boolean && boolean	boolean || boolean
true && true → true	false || false → false
true && false → false	false || true → true
false && x → false	true || x → true

! true → false

! false → true

! 5 > 2

Conditional Operator

? :

Ternary operator

Next class

Assignment Operaters

=, +=, -=, *=, /=, %=

```
x = 4;   ✓
4 = 3;   } error
4 = x;
```

```
int x = 5;
x = x + 7;
```

↑ Container ↑ Content

x += 4 → x = x + 4

x -= 7 → x = x - 7

x *= 5 → x = x * 5

Instance of Operator

object instanceof ClassName/SubClass/interface

Result will be either True or False

s1 instanceof String → true
                     → false